Trapped Ion Quantum Computing

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PHYS 411
July 2024

Big Picture

Pros

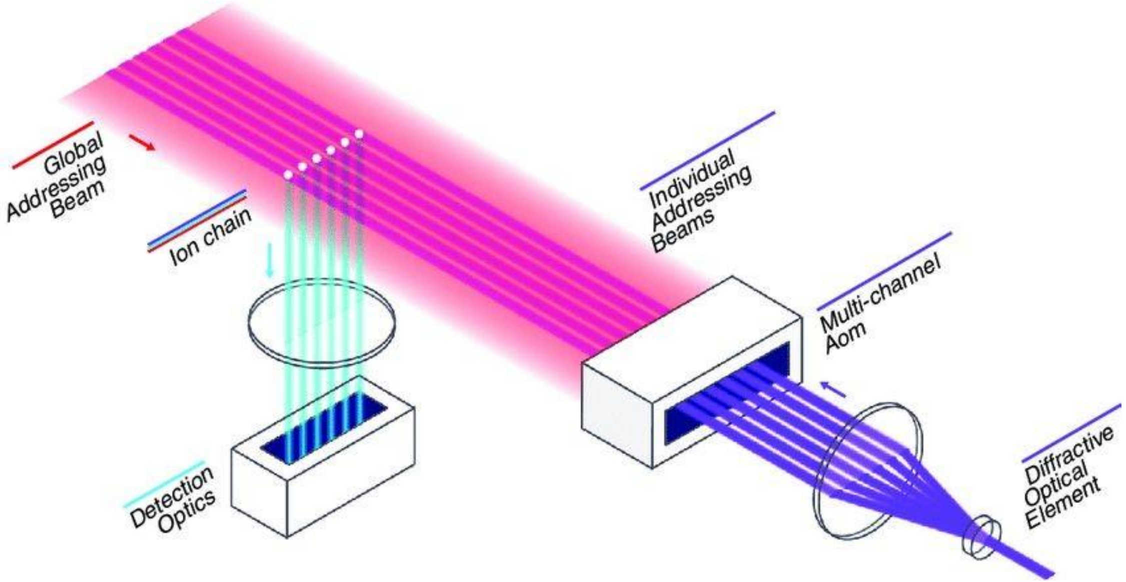
- Physical qubits (no need to invent the wheel)
- Very long coherence times
- Can operate in room temperature but in ultra high vacuum
- Highest achieved fidelities
- Very good initialization
- Single shot QND measurement
- All-to-all connectivity

Cons

- Slow gates
- Still not scalable

Nice to Have

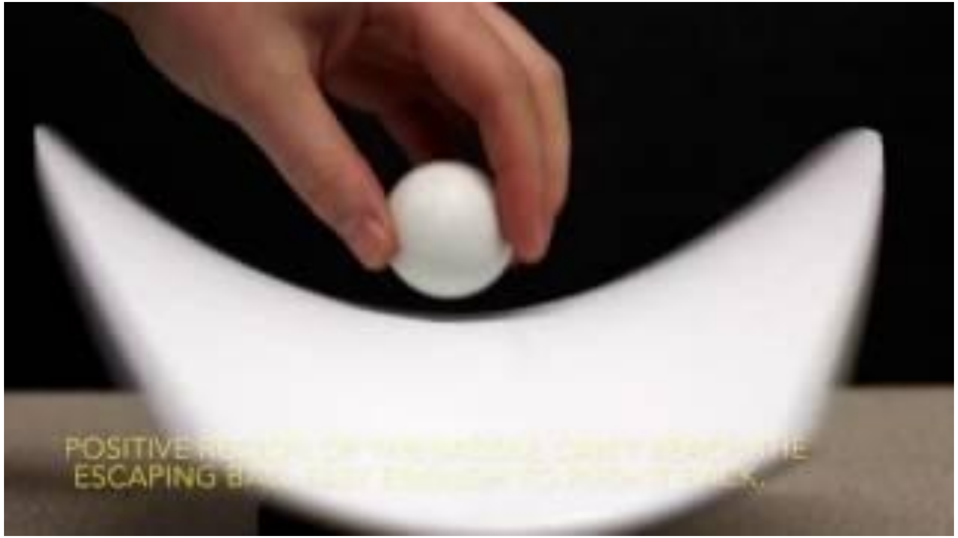
- CMOS compatibility
- Mature (turnkey) secondary equipment

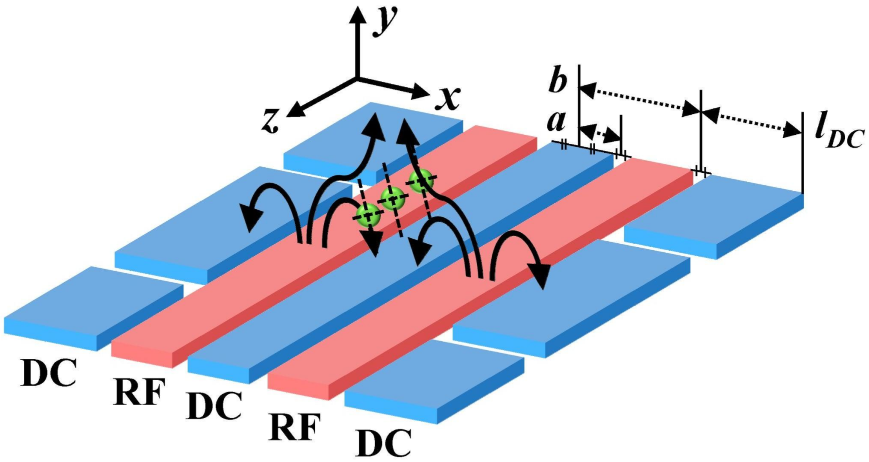


- Global beams
  - pre and post processing
- Individual addressing
  - single ion transitions

[Source: researchgate]

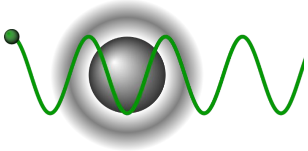
General Apparatus



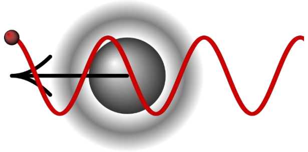


Quadrupole (aka Paul) trap

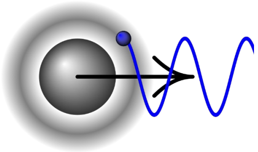
- Critical frequency -> stability regions

1


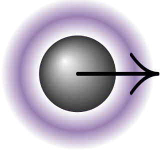
Nothing at all

2


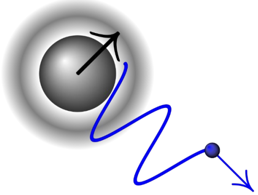
Red shifted (smaller frequency)

3.1


Blue shifted (greater frequency)

3.2


Photon absorption
Spontaneous emission in random direction

3.3


<u>repeated until equilibrium</u>

- **A laser is tuned slightly lower the transition frequency**
- **Insufficient cooling limit**

The oscillating frequency is much greater than the photon recoil one

$$\eta^2(n+1) \ll 1$$

holds of sufficient cooling

In that case the general Hamiltonian is well approximated by

$$\hat{H}_{LD}(t) = \frac{\hbar\Omega_0}{2}\hat{\sigma}_+[1 + i\eta(\hat{a}e^{-i\omega_m t} + \hat{a}^\dagger e^{i\omega_m t})]e^{i(\phi-\delta t)} + H.c.$$

$$\hat{H}_{car} = \frac{\hbar\Omega_{n,n}}{2}(\hat{\sigma}_+ e^{i\phi} + \hat{\sigma}_- e^{-i\phi})$$

δ=0

- Uncoupled -> only electronic transitions
- Used for single qubit gates and sideband cooling

$$\hat{H}_{bsb} = \frac{\hbar\Omega_{n,n+1}}{2}\eta(\hat{a}^\dagger\hat{\sigma}_+e^{i\phi} + \hat{a}\hat{\sigma}_-e^{-i\phi})$$

δ=ω

- Coupled
- Does not conserve energy

$$\hat{H}_{rsb} = \frac{\hbar\Omega_{n,n-1}}{2}\eta(\hat{a}\hat{\sigma}_+ e^{i\phi} + \hat{a}^\dagger\hat{\sigma}_- e^{-i\phi}).$$

$\delta=-\omega$

- Coupled
- Conserves energy
- Same as JC model

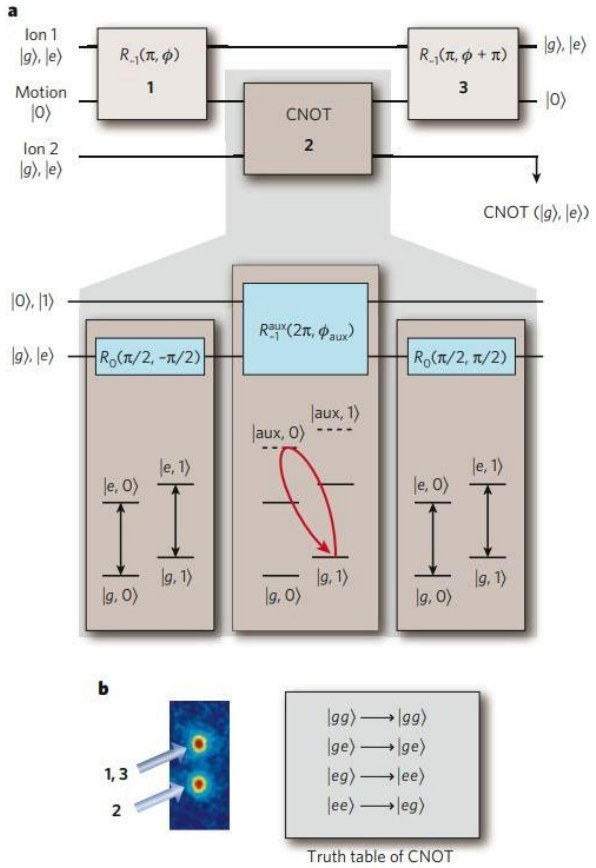
$$\hat{H}_{car} = \frac{\hbar\Omega_{n,n}}{2}(\hat{\sigma}_+ e^{i\phi} + \hat{\sigma}_- e^{-i\phi})$$

$$\hat{U}(t) = \exp[i\Omega_0 t/2\,(\hat{\sigma}_+ e^{i\phi} + H.c.)]$$

$$\hat{U}(t) = \begin{bmatrix} cos(\theta/2) & -ie^{-i\phi} sin(\theta/2) \\ -ie^{i\phi} sin(\theta/2) & cos(\theta/2) \end{bmatrix}$$

Rabi oscillation in the carrier resonance

Single qubit gates



- Two degrees of freedom
  - Vibration
  - Two level system

$$U_{m,n} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \end{bmatrix}$$

Equivalent with CX, apply Hadamard transformation

Cirac-Zoller gate